

Report: Optimizing NYC Taxi Operations

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Objective – The Analysis focuses on optimizing NYC Taxi operations with the use of 2023 taxi trip data to uncover insights that could help optimise taxi operations. The goal is to analyse patterns in the data that can inform strategic decisions to improve service efficiency, maximise revenue, and enhance passenger experience.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

The data for each month is stored in parquet file. Upon checking, each file contains around 3 million rows. If the data of all months is clubbed it would be difficult to perform analysis on population data.

Hence sampling is performed. About 1% of data is sampled randomly from the entire dataset, ensuring equal proportion of each date so that the sample data represents population data.

The sample data is stored in another parquet file. It contains about 380 K rows. The sampled data is then proceeded for cleaning.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 379268 entries, 0 to 379267
Data columns (total 20 columns):
#   Column              Non-Null Count  Dtype
---  -
0   VendorID            379268 non-null  int64
1   tpep_pickup_datetime 379268 non-null  datetime64[us]
2   tpep_dropoff_datetime 379268 non-null  datetime64[us]
3   passenger_count      366326 non-null  float64
4   trip_distance        379268 non-null  float64
5   RatecodeID          366326 non-null  float64
6   store_and_fwd_flag   366326 non-null  object
7   PULocationID         379268 non-null  int64
8   DOLocationID         379268 non-null  int64
9   payment_type         379268 non-null  int64
10  fare_amount          379268 non-null  float64
11  extra                379268 non-null  float64
12  mta_tax              379268 non-null  float64
13  tip_amount           379268 non-null  float64
14  tolls_amount         379268 non-null  float64
15  improvement_surcharge 379268 non-null  float64
16  total_amount         379268 non-null  float64
17  congestion_surcharge 366326 non-null  float64
18  airport_fee          29673 non-null   float64
19  Airport_fee          336653 non-null  float64
dtypes: datetime64[us](2), float64(13), int64(4), object(1)
memory usage: 57.9+ MB
```

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Since the data is sampled from 12 different files, the index of data is fixed and made in sequence using `reset_index()` function.

2.1.2. Combine the two `airport_fee` columns

The combination of data shows two different airport fee columns, probably due to difference in column name. Both columns are combined to form a single `airport_fee` column, by simply adding the 2 columns.

2.1.3. Fix columns with negative values

Columns with negative values are identified and fixed by replacing them with absolute values.

```
# Find which columns have negative values
non_datetime_cols = sample_data.select_dtypes(exclude=['datetime', 'object']).columns
negative_rows = sample_data[sample_data[non_datetime_cols].lt(0).any(axis=1)]

# fix these negative values
for i in sample_data.select_dtypes(exclude=['datetime', 'object']).columns :
    sample_data[i] = sample_data[i].apply(lambda x: abs(x) if x<0 else x)
```

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

The number of missing values is calculated in each column. Four columns are found to have missing values, where proportion of each being about 3.4%. These missing values are handled individually for each column.

2.2.2. Handling missing values in `passenger_count`

The missing values in passenger count column is handled by replacing with most occurring value. The missing data is replaced by 1.
The rows with 0 passenger count are removed.

2.2.3. Handle missing values in `RatecodeID`

The missing values in 'RatecodeID' column are replaced by '1' – which comprise about 95% of counts.

2.2.4. Impute NaN in `congestion_surcharge`

The congestion surcharge is either 0 or 2.5. All missing values are currently filled with 0, later while handling outliers these values are fixed further.

2.2.5. Handle missing values in `store_and_fwd_flag`

The missing values in `store_and_fwd_flag` columns are filled by 'N' which maintained the proportion of values in the column.

VendorID	0.000000	VendorID	0.0
tpep_pickup_datetime	0.000000	tpep_pickup_datetime	0.0
tpep_dropoff_datetime	0.000000	tpep_dropoff_datetime	0.0
passenger_count	3.412363	passenger_count	0.0
trip_distance	0.000000	trip_distance	0.0
RatecodeID	3.412363	RatecodeID	0.0
store_and_fwd_flag	3.412363	store_and_fwd_flag	0.0
PULocationID	0.000000	PULocationID	0.0
DOLocationID	0.000000	DOLocationID	0.0
payment_type	0.000000	payment_type	0.0
fare_amount	0.000000	fare_amount	0.0
extra	0.000000	extra	0.0
mta_tax	0.000000	mta_tax	0.0
tip_amount	0.000000	tip_amount	0.0
tolls amount	0.000000	tolls amount	0.0

2.3. Handling Outliers and Standardizing Values

2.3.1. Outliers in trip distance

From trip distance column following data was found and dropped-

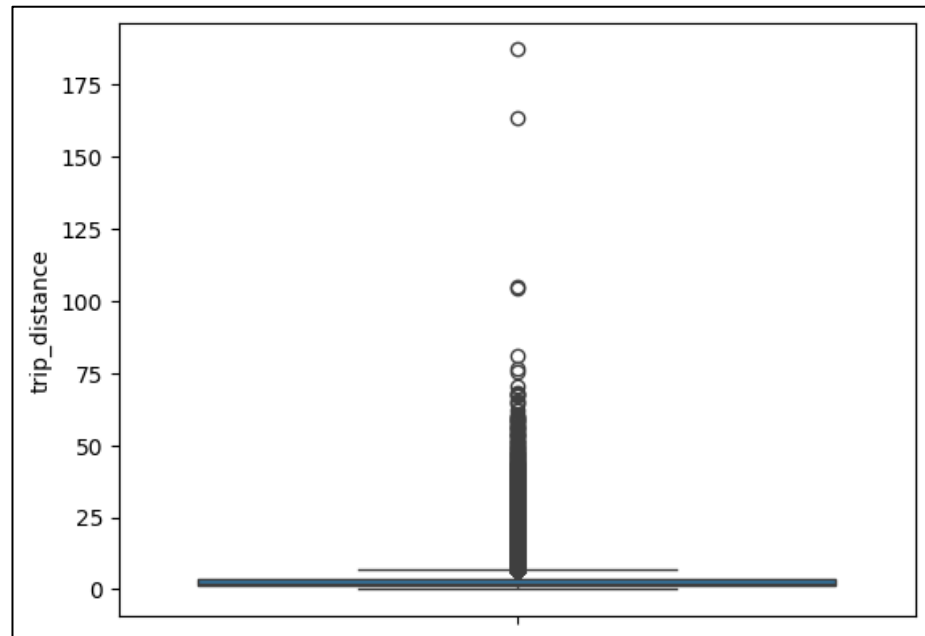
- Trip distance is 0 but fare amount is not zero
- Trip distance is 0, pickup and drop-off locations are same but fare amount is not zero
- Trip distance is more than 200 miles

We further observe the trip distance column and check for potential outliers. Upon visualizing the column, we check further to remove trip distance more than 100 miles, if they are considered as outliers to be removed.

```
sample_data['trip_distance'].describe()
✓ 0.0s
```

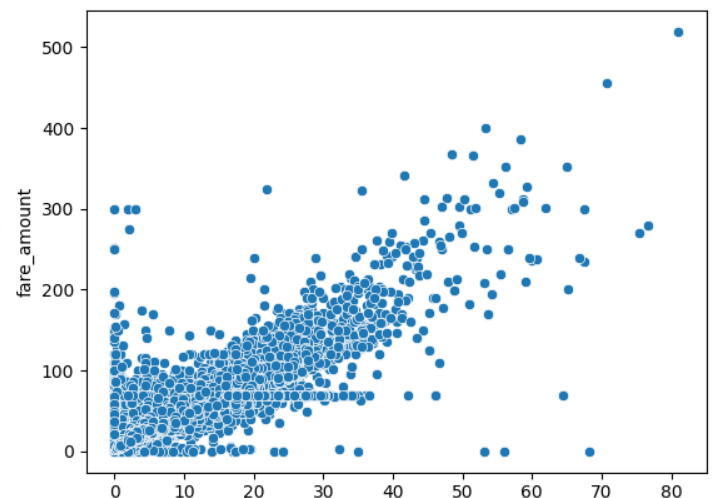
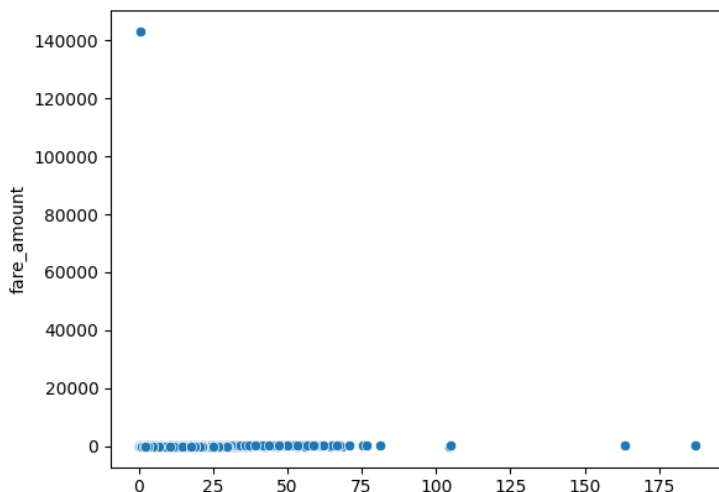
count	370449.000000
mean	3.473016
std	4.542667
min	0.000000
25%	1.070000
50%	1.800000
75%	3.420000
max	187.350000

Name: trip_distance, dtype: float64



2.3.2. Outliers in fare amount

- Upon checking fare amount with trip distance, fare amount more than \$ 1000 and trip distance more than 100 miles are dropped. While still some outliers are observed, there could be other factors to determine the variation.



2.3.3. Outliers in payment type

- Rows with payment type 0 and 0 trip distance are dropped.
- For remaining rows with 0 payment type, the rows are assigned payment type 1 or 2 in the proportion of 5:1, to maintain the ratio of payment types in the data.

payment_type	
1	0.796070
2	0.166591
0	0.027190
4	0.006754
3	0.003395

```
sample_data = sample_data.reset_index(drop=True)
p_0 = sample_data[sample_data['payment_type']==0]
num = p_0.shape[0]

num_1= int(round(num*(5/6),0))
num_2=int(round(num*(1/6),0))

fill_values = np.array([1] * num_1 + [2] * num_2)
np.random.shuffle(fill_values)
sample_data.loc[p_0.index, 'payment_type'] = fill_values

sample_data['payment_type'].value_counts(normalize =True)
```

payment_type	
1	0.818729
2	0.171123
4	0.006754
3	0.003395

2.3.4. Outliers in mta_tax

- The mta tax value needs to be 0.5 or 0, we simply replace other occurring values with 0.5

2.3.5. Imbalanced total amount value

- We observed imbalance in total amount, which could have affected other columns such as airport charges, congestion surcharge and mta tax values.
- The difference between given total amount and actual total amount is calculated and mentioned columns are fixed accordingly, thus reducing error in analysis

2.3.6. Outliers in airport fee column

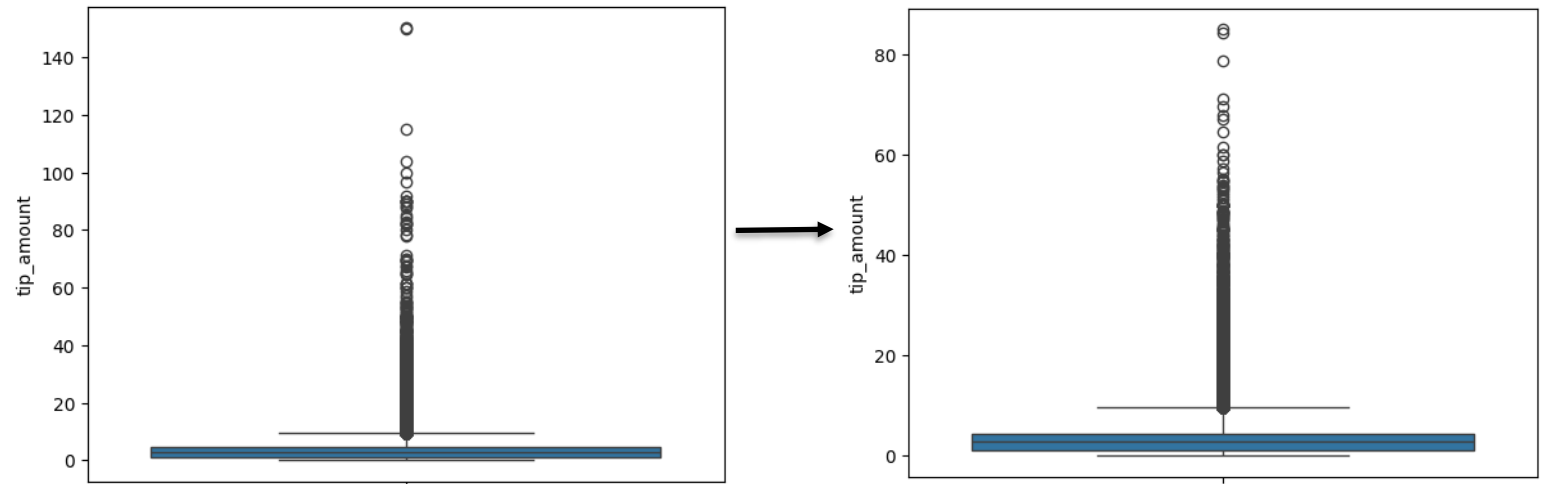
- Airport fee is either 0 or \$ 1.25, values 1.75 are replaced by 1.25

2.3.7. Outliers in RateCodeID column

- The RateCodeID are categorical data between 1 to 6. The value 99 is anyway small in proportion, yet replaced with most occurring value of 1.

2.3.8. Outliers in tip amount columns

- Tip amount for more than \$50 looks unlikely to occur. Upon closely observing the data, the trip distance comes out to be less than 5 miles, which declares them outliers.
- Rows with subsequent data is removed.
- For some tip amounts more than 50 , the tip amount looks justified against trip distance



3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

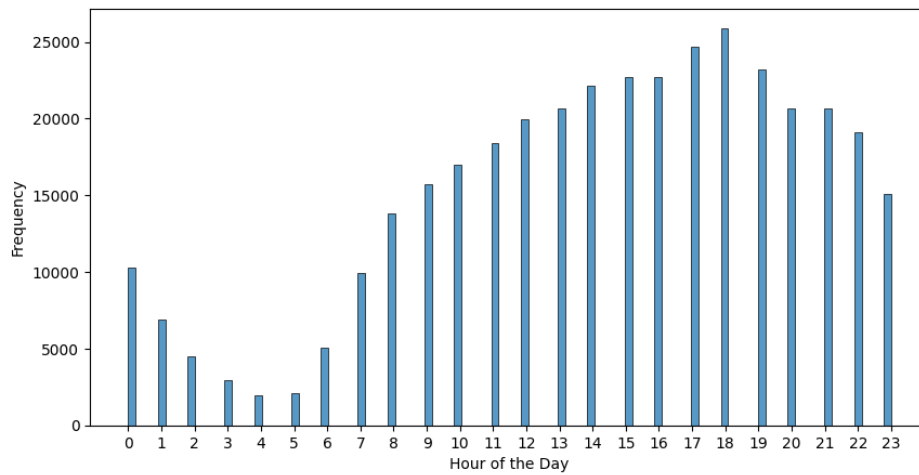
3.1.1. Classify variables into categorical and numerical

The variables are classified as numerical and categorical as follows:

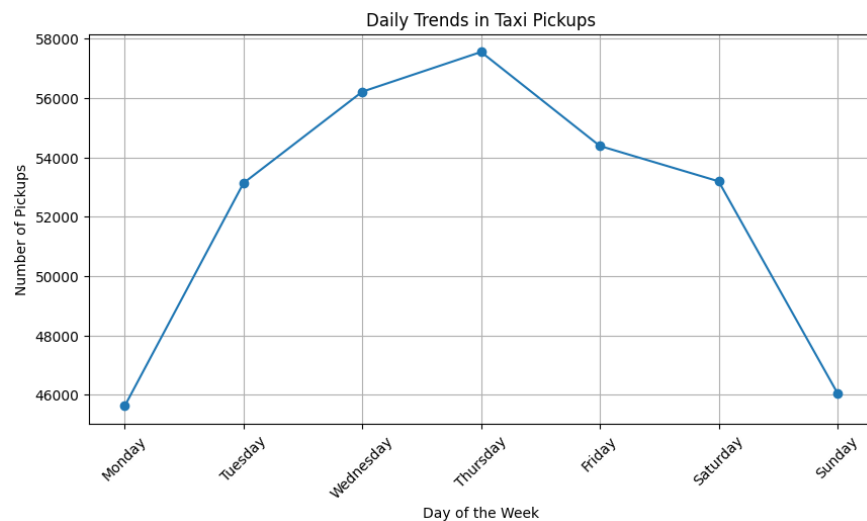
'VendorID'	Numerical
'tpep_pickup_datetime'	Numerical
'tpep_dropoff_datetime'	Numerical
'passenger_count'	Numerical
'trip_distance'	Numerical
'RatecodeID'	Categorical
'store_and_fwd_flag'	Categorical
'PULocationID'	Categorical
'DOLocationID'	Categorical
'payment_type'	Categorical
'fare_amount'	Numerical
'extra'	Numerical
'mta_tax'	Numerical
'tip_amount'	Numerical
'tolls_amount'	Numerical
'improvement_surcharge'	Numerical
'total_amount'	Numerical
'congestion_surcharge'	Numerical
'airport_fee'	Numerical

3.1.2. Analyze the distribution of taxi pickups by hours, days of the week, and months

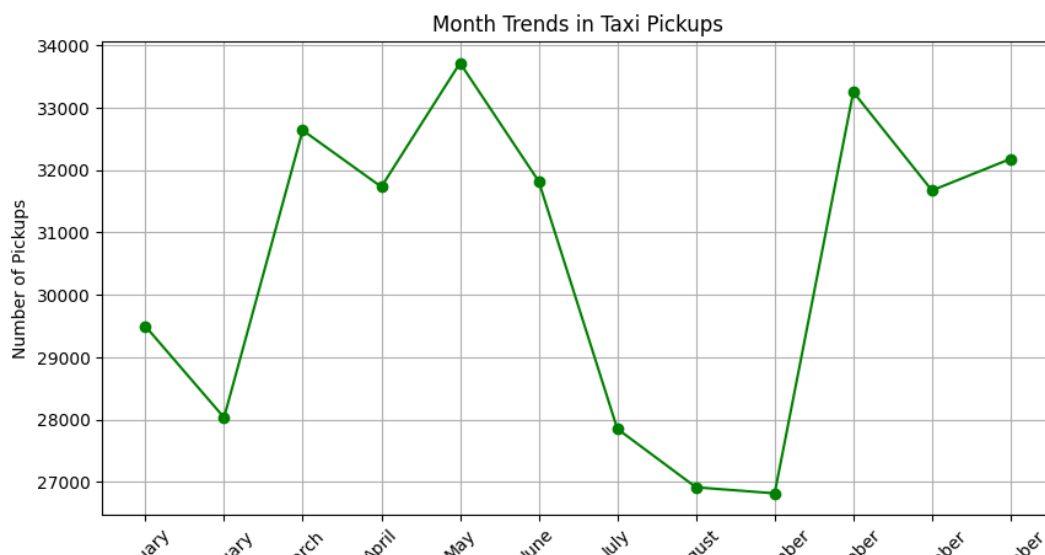
- The hourly trend shows the greatest number of cabs taken during evening time, around 4-7 PM as it is the peak office leave hour. The least taxis are taken during early hours of the day – 3AM -5 AM



- The day trend most cabs are taken on Thursday. Data Surprises with least number of pickups on Monday



- Monthly trend shows dip in number of pickups during the month of monsoon- July to September, with sudden shoot in month of October.

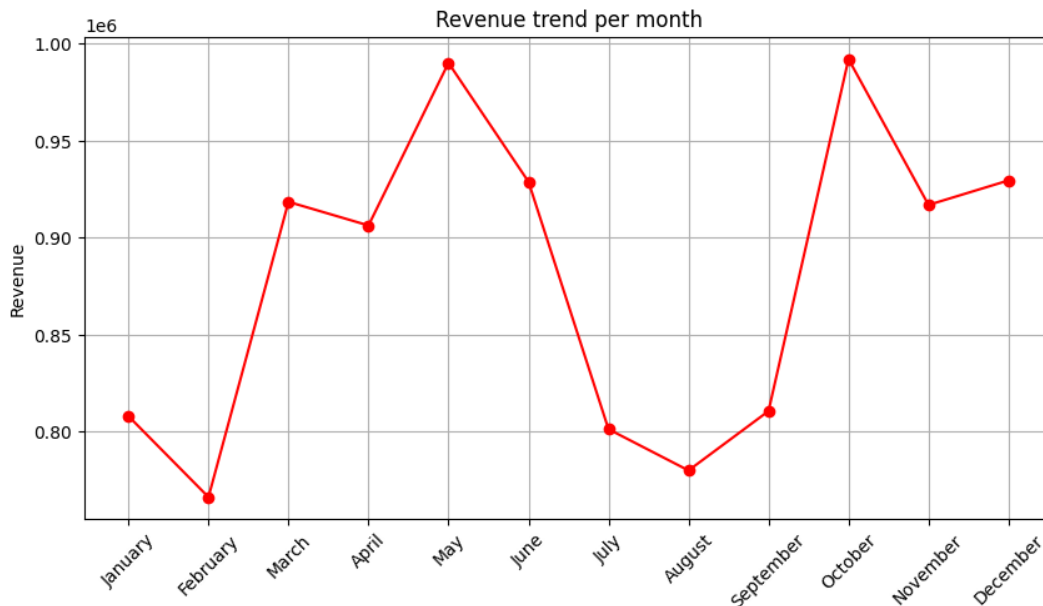


3.1.3. Filter out the zero/negative values in fares, distance and tip

- There are some rows which contains instant cancelled rides- where fare amount, trip distance, tips – all are zero. We find these rows and filter them, as they add no value to the analysis and are considered additional non-valued data.
- Trips with same pickup and drop zones and zero trip distance are also filtered, contains no data to consider

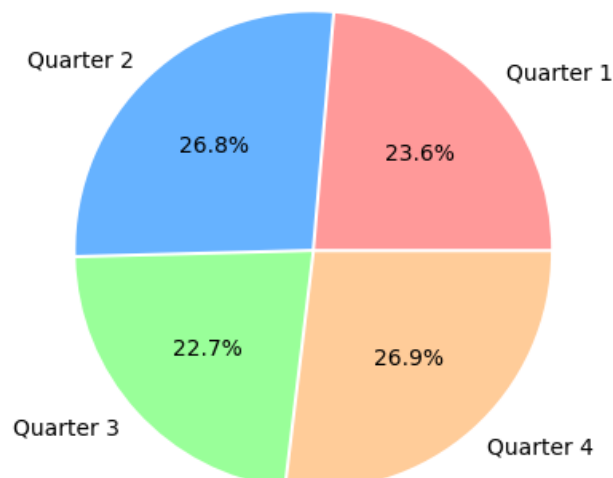
3.1.4. Analyse the monthly revenue trends

Monthly revenue trend looks like following number of pickups, as more the number of rides more the revenue.



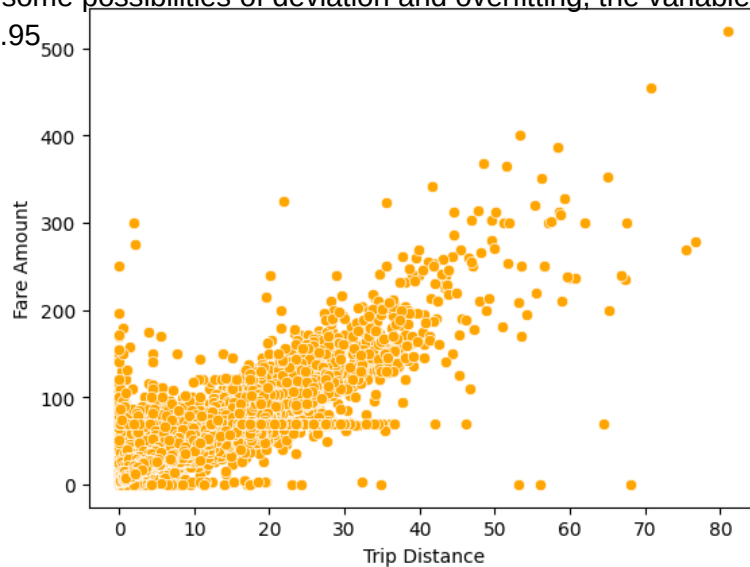
3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

Revenue collected in each quarter is almost same, yet slightly higher in Quarter 2 and Quarter 4



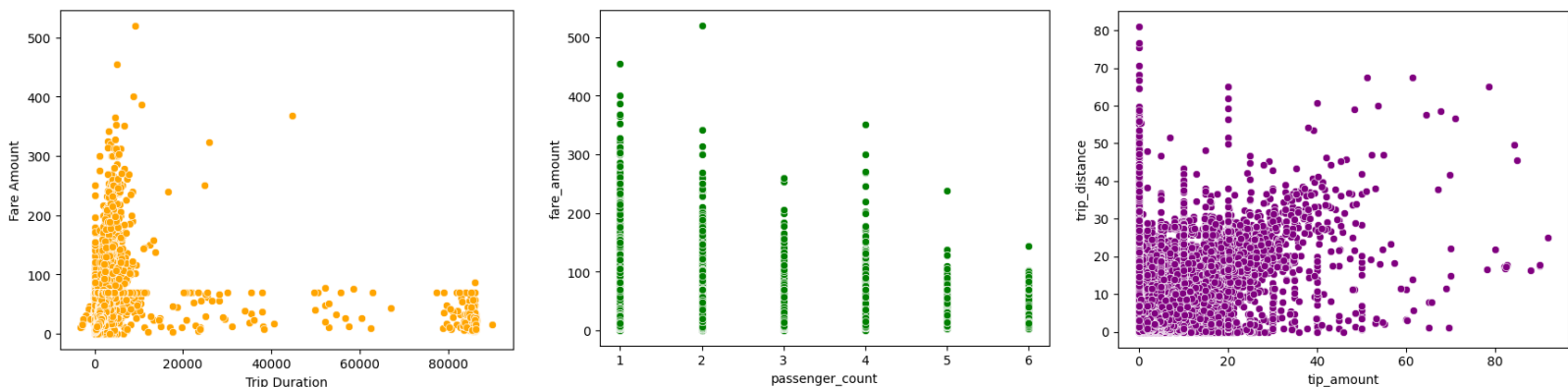
3.1.6. Analyse and visualise the relationship between distance and fare amount

A strong linear relation is observed between trip distance and fare amount. While there are some possibilities of deviation and overfitting, the variables show strong correlation of 0.95



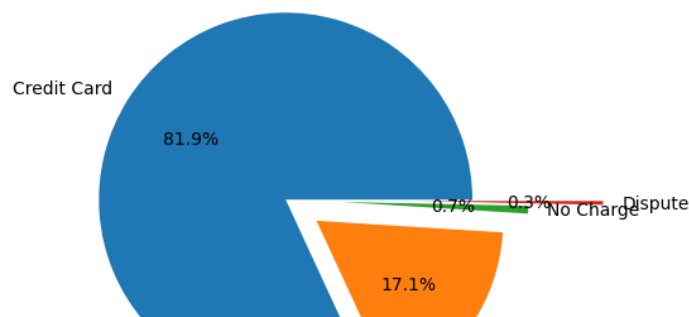
3.1.7. Analyze the relationship between fare/tips and trips/passengers

- Not much of strong relation between fare amount with trip duration is established. The correlation factor comes out to be 0.28
- No relation between fare amount and passenger count is found
- Tip amount is slightly related to trip distance. Some high tip amount are observed for short distances as well, assuming exceptional cases

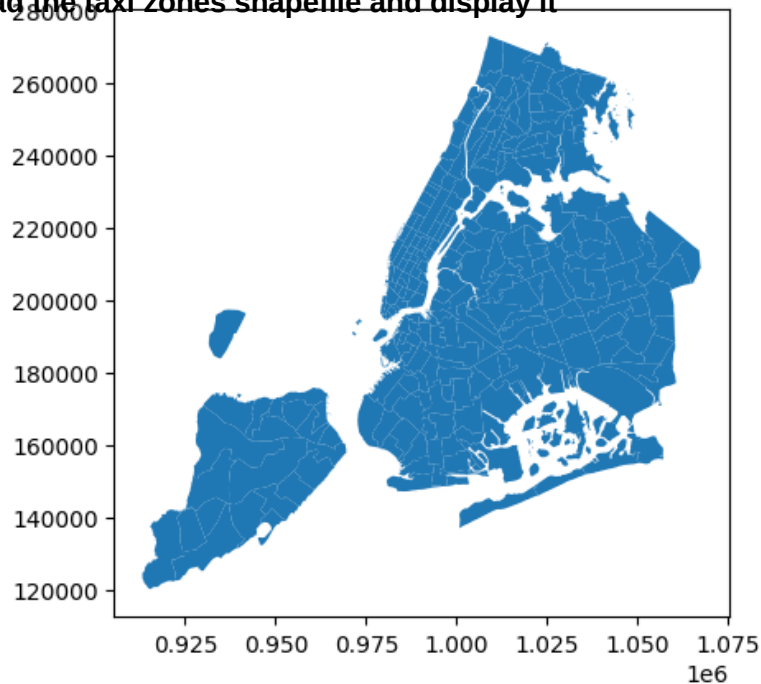


3.1.8. Analyse the distribution of different payment types

- About 82% of payment are done through credit card; 17% are done through cash. No Charge and dispute collectively contributes 1% of total transactions.



3.1.9. Load the taxi zones shapefile and display it



3.1.10. Merge the zone data with trips data

```
# Merge zones and trip records using locationID and PULocationID
sample_data2= pd.merge(sample_data,zones,left_on='PULocationID',right_on='LocationID')
```

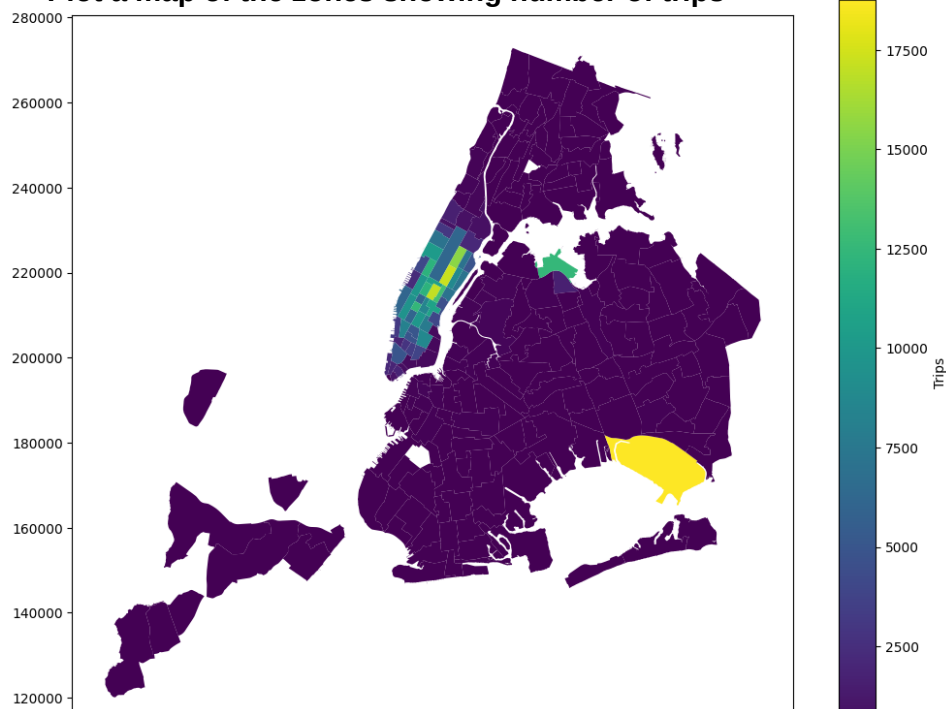
3.1.11. Find the number of trips for each zone/location ID

```
# Group data by location and calculate the number of trips
location_trips =pd.DataFrame(sample_data2['LocationID'].value_counts()).reset_index()
```

3.1.12. Add the number of trips for each zone to the zones dataframe

```
# Merge trip counts back to the zones GeoDataFrame
zones= pd.merge(zones,location_trips, left_on = 'LocationID',right_on = 'LocationID')
```

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

- The analysis shows busiest hours being evening hours 4- 6PM, busiest day being Thursday while busiest months being November-December
- The revenue collected have been the most in the month of May and October, while least in February
- Fare Amount shows strong linear relationship with trip distance. However, other factors such as trip duration, drop location may contribute in higher fare amount
- As we observe the zone map, the airport zones and area around Central Park have been the busiest zones.

3.2. Detailed EDA: Insights and Strategies

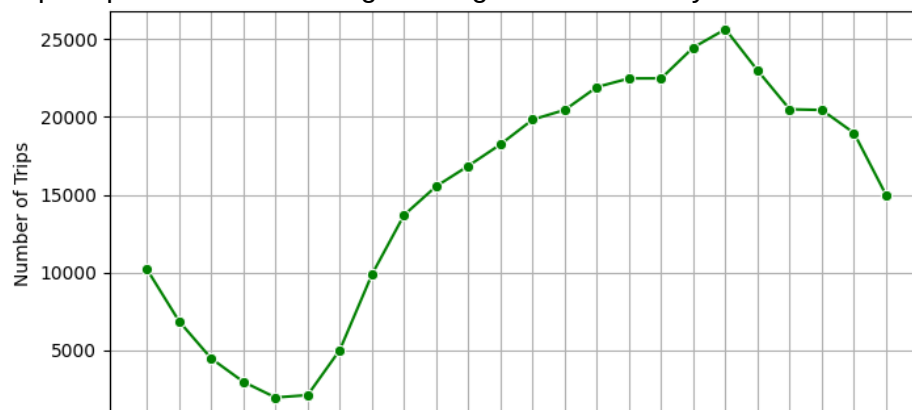
3.2.1. Identify slow routes by comparing average speeds on different routes

- The slow route is identified for every hour of the day. Identifying such high traffic routes indicates high demand which shall help to divert more taxis towards that area. Due to high travel time extra cost is applicable which further increases the revenue

Hour	LocationID	speed
0	116	7.292450
1	134	7.769784
2	89	0.303030
3	82	8.325188
4	18	1.199001
5	219	0.882137
6	70	0.085750
7	128	0.115385
8	134	1.106331
9	257	1.487603
10	26	3.092784
11	1	0.307692
12	169	2.241840
13	53	2.658789
14	169	2.184697
15	127	0.318021
16	91	1.285059
17	196	0.830450
18	131	2.100128
19	3	0.529023
20	91	4.749340
21	47	1.939394
22	129	6.465066
23	67	0.723861

3.2.2. Calculate the hourly number of trips and identify the busy hours

- Evening hours between 4 PM – 7PM are the busiest hours of the day. The number of pickups is the most during evening hours of the day.



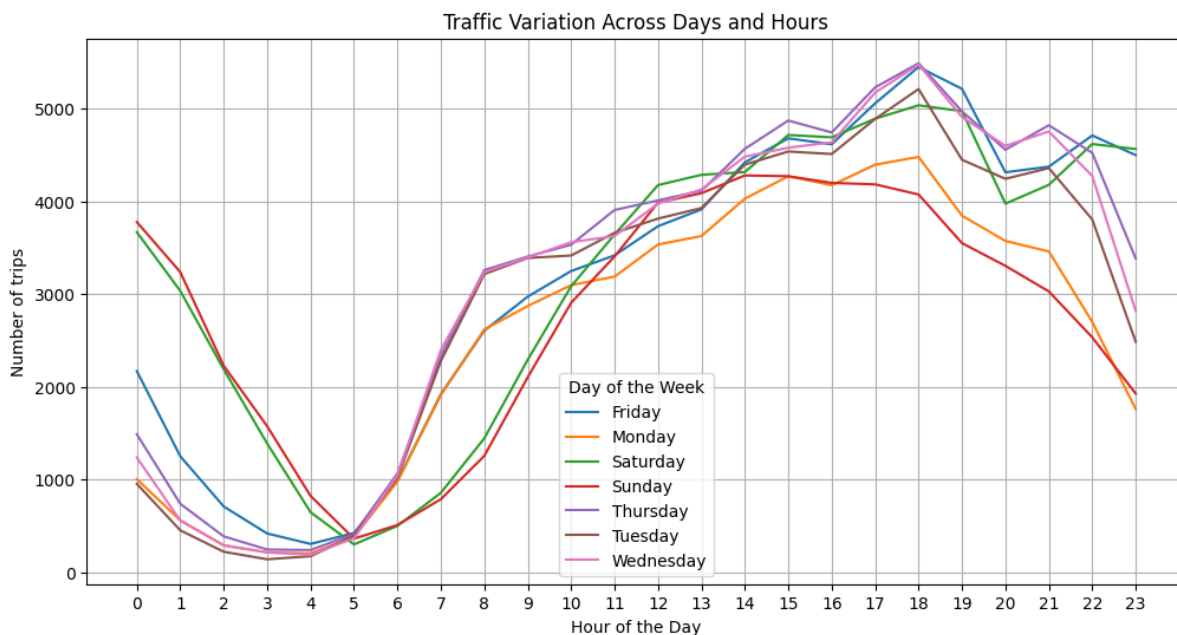
3.2.3. Scale up the number of trips from above to find the actual number of trips

- As the analysis is performed on a sample, by taking 1% of total database, the actual number of trips can be estimated by scaling with sampling factor.
- The five busiest hours comes up to be evening hours

18	2564900
17	2446900
19	2299400
16	2248500
15	2248300

3.2.4. Compare hourly traffic on weekdays and weekends

- The cab rides on weekend are observed more than weekdays during early hours of the day. This could be due to flight schedules.
- Evening rides on Saturdays are more than that of Sundays.
- On weekdays, rides on Monday evening are least as compared to other weekdays.



3.2.5. Identify the top 10 zones with high hourly pickups and drops

- Airports are within top 10 high hourly pickup zones. That sounds promising as airports are busiest zones of pickups.
- As compared to average hourly pickups, average hourly drops are lesser.

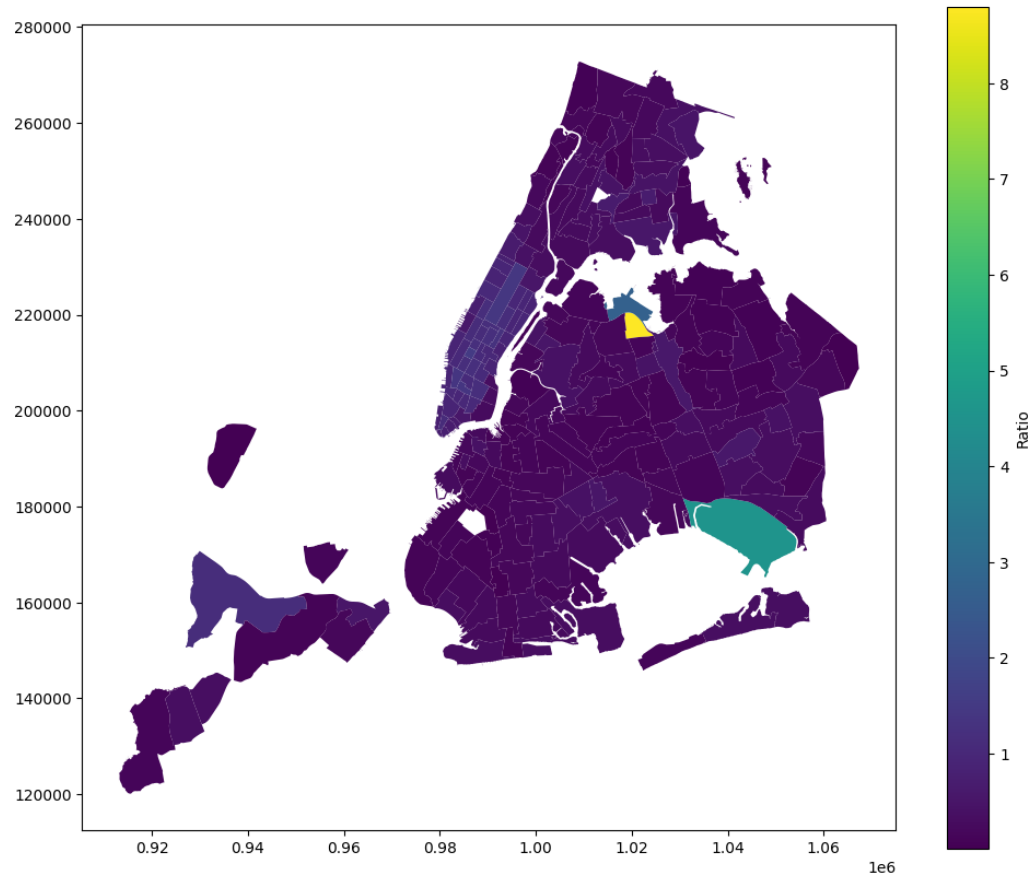
zone	Avg hourly pickup
JFK Airport	782.041667
Upper East Side South	720.500000
Midtown Center	708.875000
Upper East Side North	647.000000
Midtown East	545.625000
LaGuardia Airport	522.250000
Penn Station/Madison Sq West	519.166667
Lincoln Square East	504.000000
Times Sq/Theatre District	503.000000
Upper West Side North	493.000000

zone	Avg hourly dropoff
Upper East Side North	680.541667
Upper East Side South	644.875000
Midtown Center	596.416667
Times Sq/Theatre District	462.791667
Murray Hill	453.333333
Midtown East	434.000000
Lincoln Square East	430.791667
Upper West Side South	423.875000
Upper West Side North	400.000000
Upper West Side West	400.000000

3.2.6. Find the ratio of pickups and dropoffs in each zone

zone	PU_DO_ratio
East Elmhurst	8.80
JFK Airport	4.56
LaGuardia Airport	2.69
Penn Station/Madison Sq West	1.53
Central Park	1.39
Greenwich Village South	1.36
West Village	1.36
Midtown East	1.26
Union Sq	1.19
Midtown Center	1.19

zone	PU_DO_ratio
Roosevelt Island	0.06
Belmont	0.06
Bayside	0.06
Whitestone	0.05
Glendale	0.05
Windsor Terrace	0.03
Ocean Parkway South	0.03
Glen Oaks	0.03
Queensboro Hill	0.02
Newark Airport	0.01

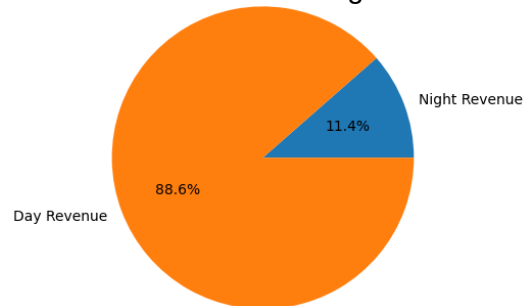


3.2.7. Identify the top zones with high traffic during night hours

zone	PU_DO_ratio
JFK Airport	11.12
East Elmhurst	6.62
LaGuardia Airport	5.42
Greenwich Village South	3.48
West Village	2.76
Flushing Meadows-Corona Park	2.33
Lower East Side	2.26
Midtown Center	2.20
Westchester Village/Unionport	2.00
Times Sq/Theatre District	1.95

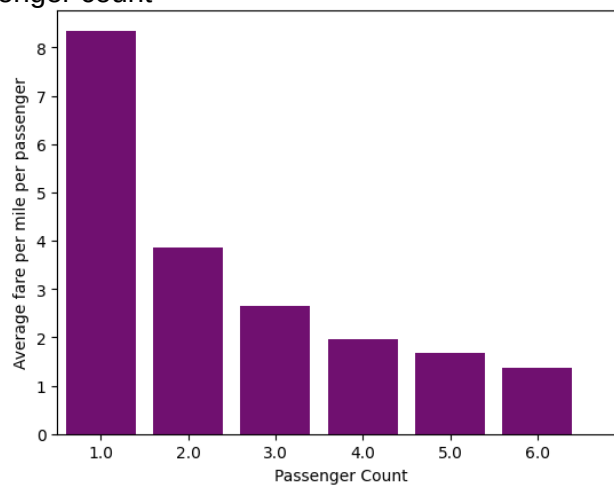
3.2.8. Find the revenue share for nighttime and daytime hours

- The night hours are considered between 23:00 hrs to 05:00 hrs. Revenue collected in daytime is almost 7.7 times of nighttime collection.



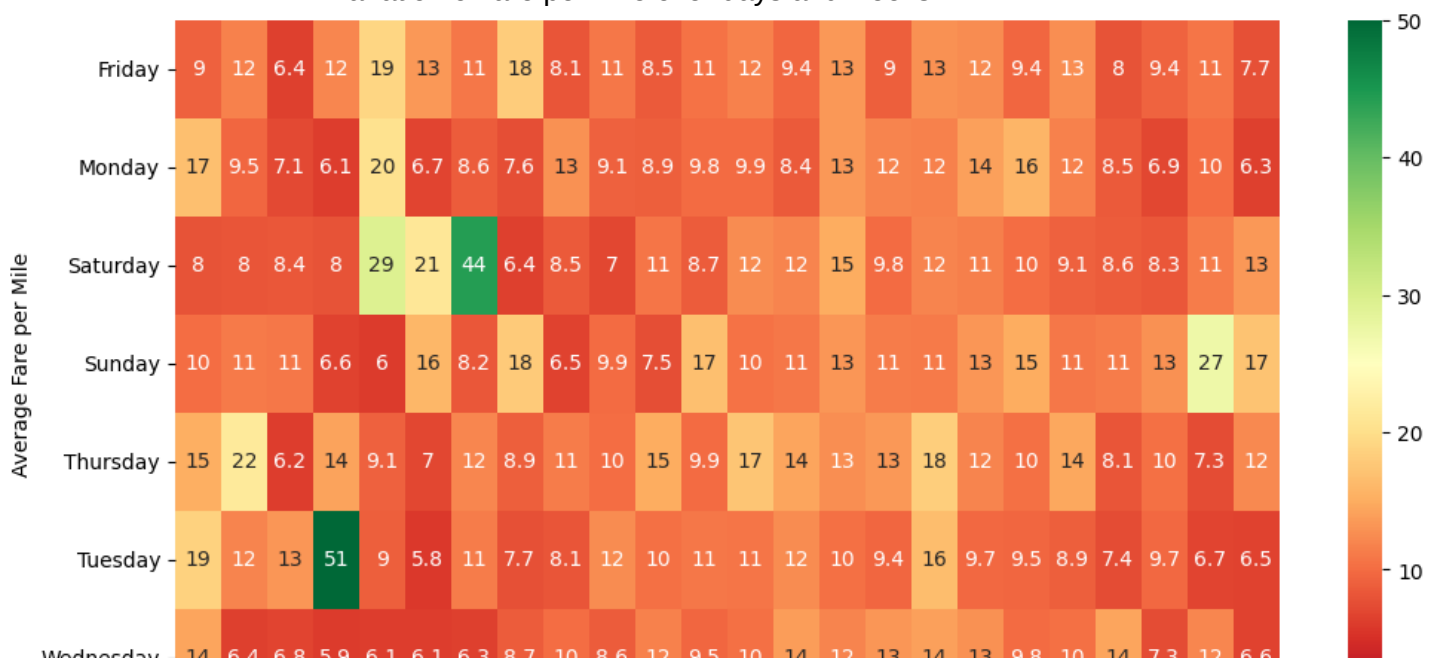
3.2.9. For the different passenger counts, find the average fare per mile per passenger

- The average fare per mile per passenger certainly decreases with increase in passenger count



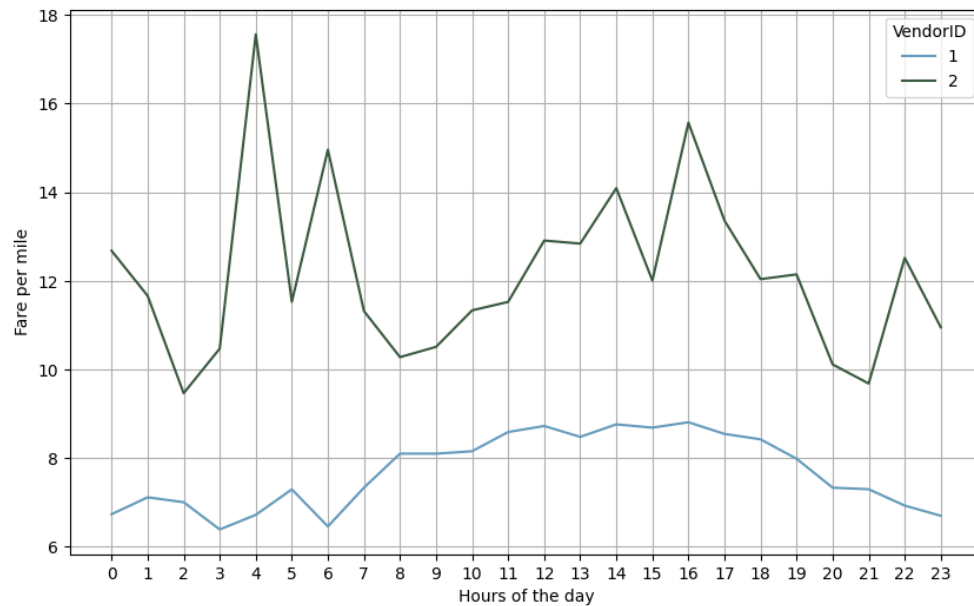
3.2.10. Find the average fare per mile by hours of the day and by days of the week

- The heatmap shows comprehensive relation between the parameters and reveal variation of fare per mile over days and weeks



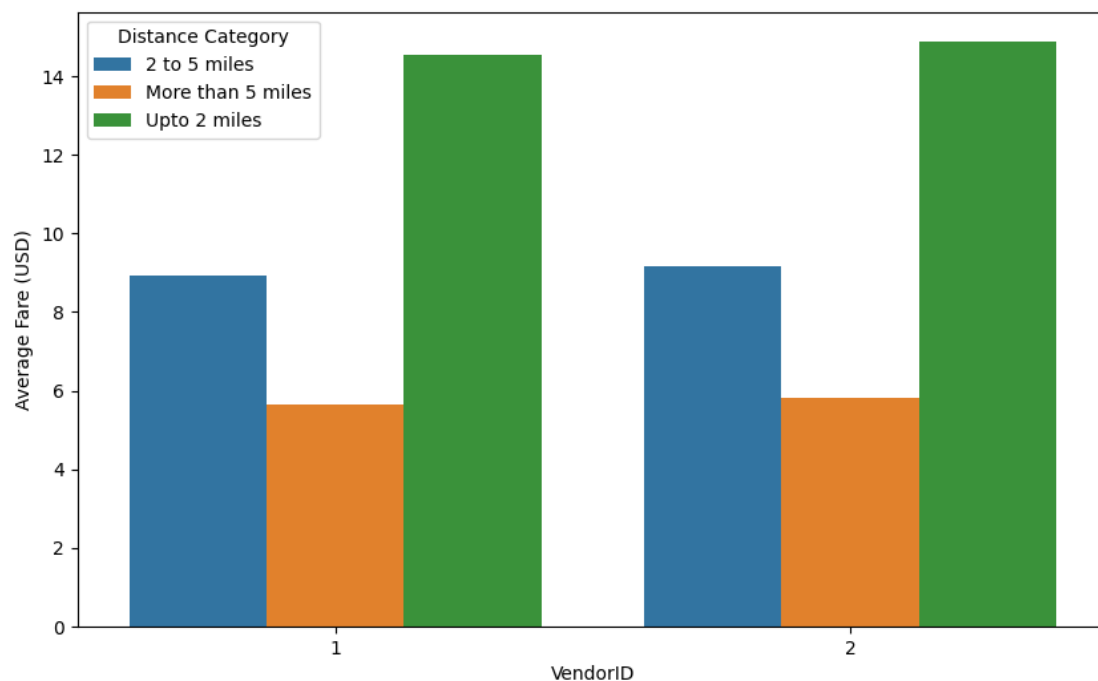
3.2.11. Analyze the average fare per mile for the different vendors

- The price range of Vendor 1 is lower than the price range of Vendor 2. While the variation is observed for Vendor 2, it is relatively smoother for Vendor 1.



3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

- The distance travelled is categorized into 3 parts – Upto 2 miles, Between 2 to 5 miles and more than 5 miles.
- The Average fare is similar for both the vendors, yet slightly higher for Vendor 2. As we saw fare per mile is more for Vendor 2, yet categorical fare rates are similar.
- The fare rate upto 2 miles is at higher side than other categories.

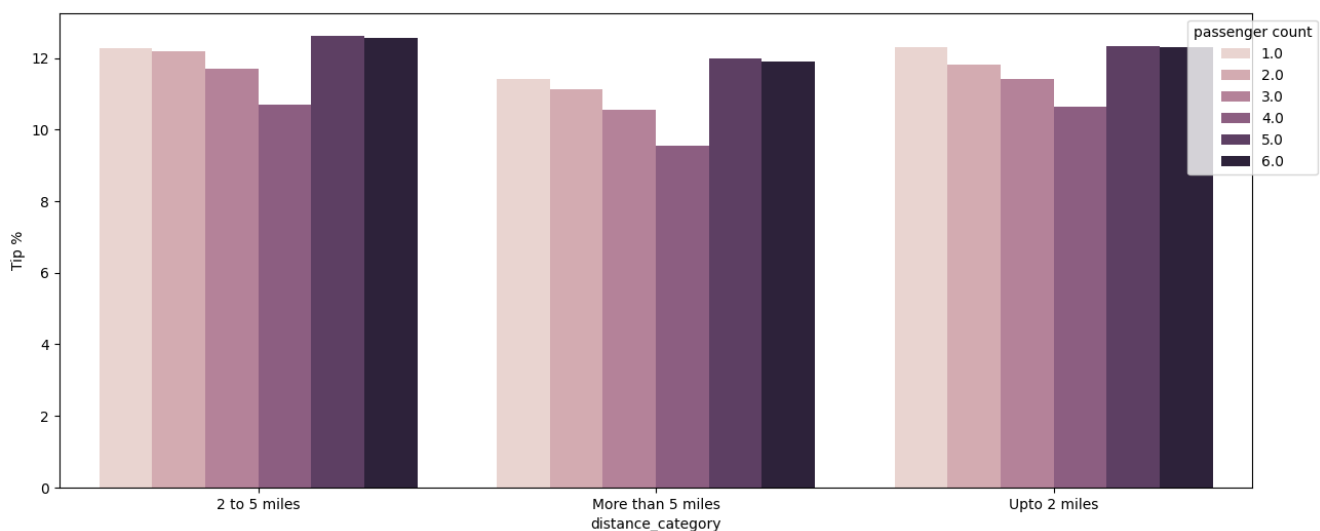


3.2.13. Analyse the tip percentages

- Tip Percentages are nearby around 12% of fare-amount. The tip percentage are comparatively lower for trips more than 5 miles.

distance_category	
2 to 5 miles	12.216336
More than 5 miles	11.302257
Upto 2 miles	12.158921

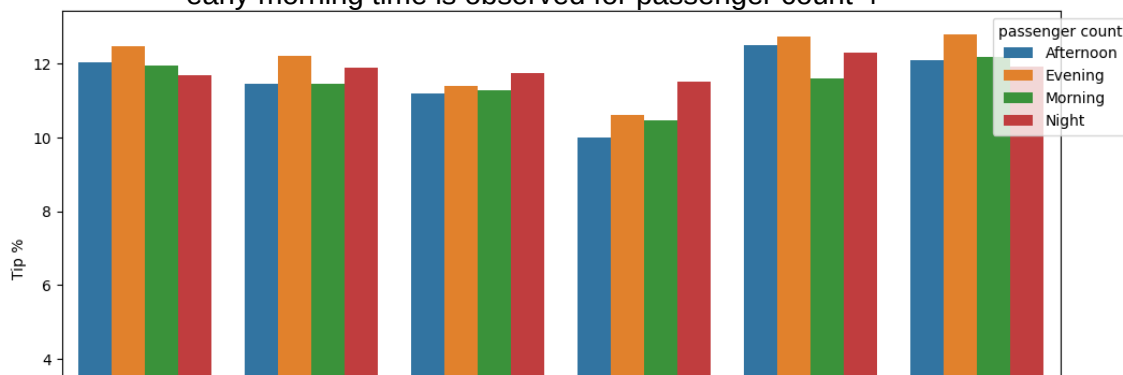
- Tip % are lower than other categories for more than 5 miles of distance category.



- The hours of days are categorized for better analysis:
Morning: 5am-12pm, afternoon: 12pm- 6pm, evening:6pm-00 am, night: 00am-5am

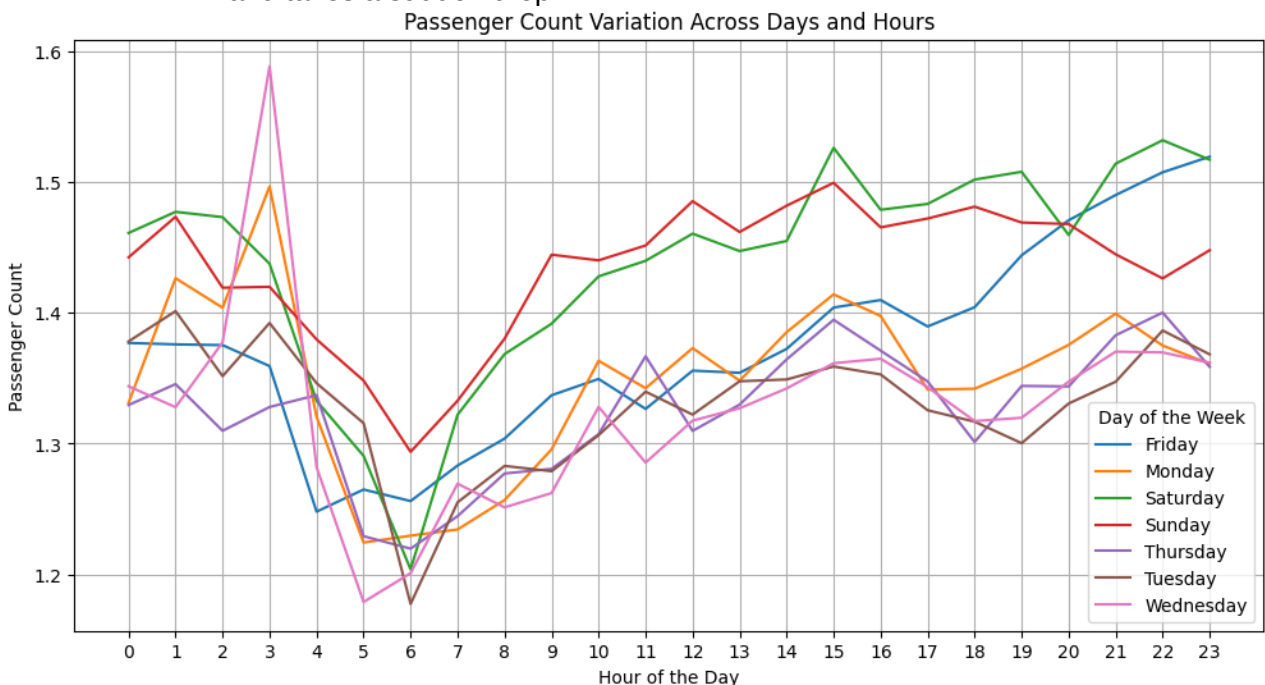
time_of_day	distance_category	
Afternoon	2 to 5 miles	11.999284
	More than 5 miles	11.311298
	Upto 2 miles	11.992538
Evening	2 to 5 miles	12.682861
	More than 5 miles	11.683722
	Upto 2 miles	12.394467
Morning	2 to 5 miles	11.828360
	More than 5 miles	10.922821
	Upto 2 miles	12.143465
Night	2 to 5 miles	11.968014
	More than 5 miles	10.787862
	Upto 2 miles	12.007359

- Tip amount remains same for passenger count. Although a bit dip during night and early morning time is observed for passenger count 4



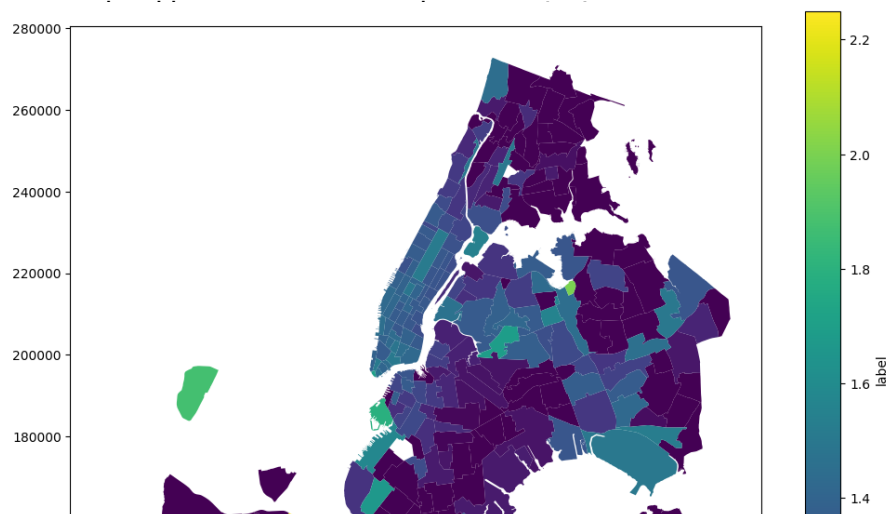
3.2.14. Analyse the trends in passenger count

- The average passenger count variation across hours of day per day of week is shown as line graph
- The average passenger count varies between 1.2 to 1.5 passengers per trip .Passenger count rises the highest during weekends, with lowest during weekdays evenings
- Wednesday shows significant variation, as passenger count is highest at 03:00 hrs and takes a sudden drop.



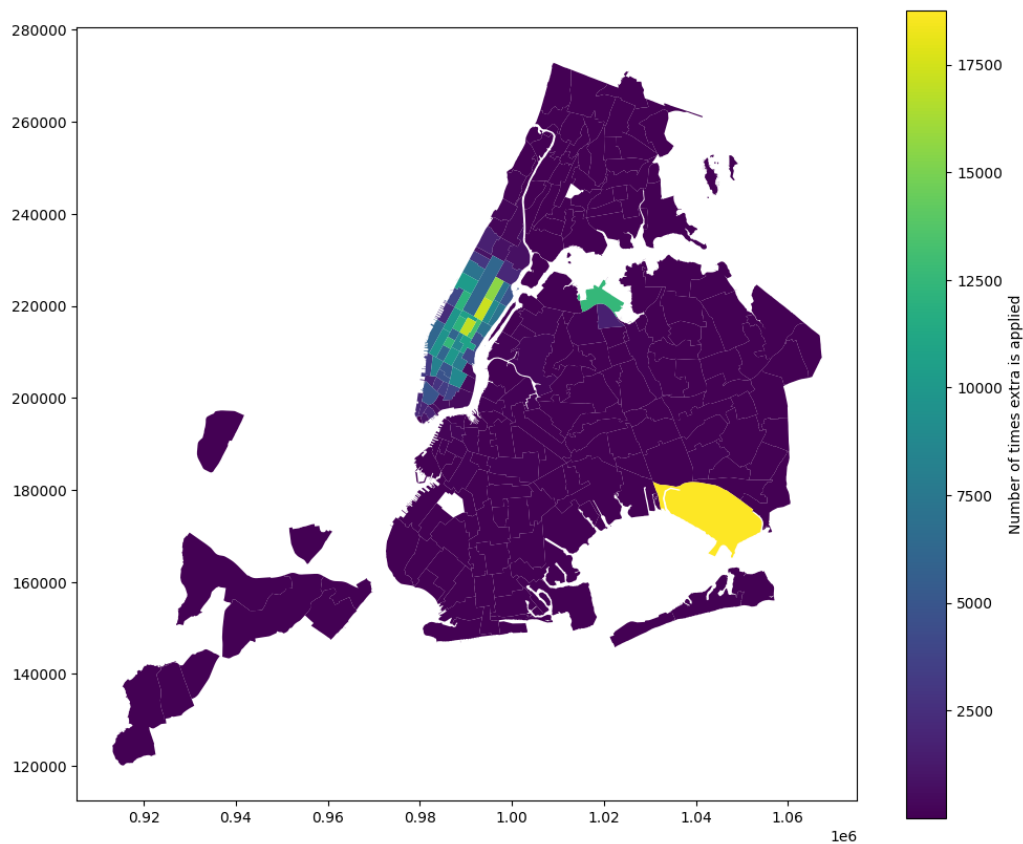
3.2.15. Analyse the variation of passenger counts across zones

- We surprisingly saw high average passenger count at brighton heights. The reason could be tourist or holiday destination.
- Aiport zones observe to have average of 1.6 passenger count, which shows more

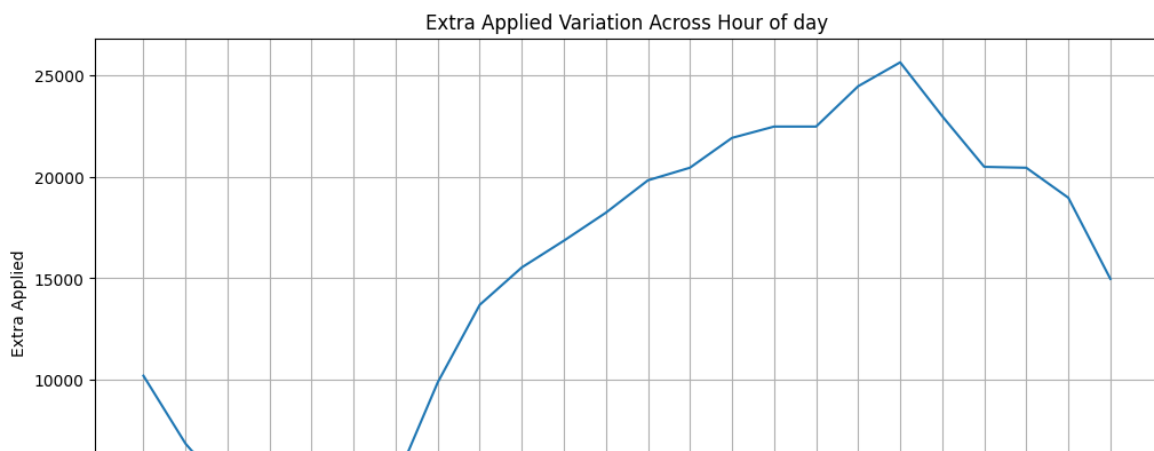


3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

- Extra charges, congestion surcharge and improvement surcharge are considered as extra charges applied.
- If we observe zones where extra charges are applied more frequently, John F Kennedy Airport and area around Central Park shows a greater number of trips.



- Peak traffic hours are deterministic factor of frequency of extra charges application. The charges occur more during evening hours, where several traffic factors increase.



4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. **Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.**

- To improve demands of taxis in high-demand areas, the drop-offs at high pickup-drop-off ration zones could be accepted more frequently.
- Low pickup-drop-off ratio could also mean more drops than pick-ups. Offbeat places like Richmond hills or forest hills trips could be accepted more with reduced extra charges to increase demand at these places.
- During morning hours taxis can be deployed to more traffic areas such as zones near airports or central park or time square areas, to meet increase in demand during peak hours.

4.1.2. **Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.**

- When observed the pickup and drops at different zones, spots like time square, central park, bay area and most important airports see most traffic and pickups.
- During evening hours the demand and traffic rises, hence more taxis can be placed at these locations to fulfil rise in demands
- Airports provide us long distance rides, and fare amount is strongly correlated to distance. Hence, a taxi booth or pickup zone could be established.
- Weekends and Tuesdays are observed to get a greater number of bookings. In order to increase rides during other days, taxis could be deployed more in uptown areas.

4.1.3. **Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.**

- In order to increase revenue during months of winter, average price for distance more than 5 miles can be increased, by \$1 -\$1.5 per mile, it can be checked to not apply surcharge amount upto certain distance to keep pricing comparable.
- Prices can be increased by 25% during peak hours and during weekends, as demand rises
- We observe lesser average fair price per mile during Friday evening hours. These can be increased by 10% to ensure increase in revenue.
- To improve night revenue factor, cabs could be deployed to areas where extra charges are applied more.

- Vendors have equal average fare prices, with price range of vendor 1 lesser than vendor 2. Vendor 1 can hence be deployed more at high pickup zones to ensure optimum pricing, meet demand and hence increase revenue.
- The low average rate per mile could be due to insufficient demand. Discount offers could be offered for longer distance rides to ensure rise in demand and thus increase pricing for the routes.